

Oversampling

UART RX cannot rely on one sample per bit, the line transitions are noisy and slow

Starter sixteen times

- Starter preset: sixteen times oversampling of a clean zero bit cell, ideal skew zero
- Sixteen sample ticks numbered zero through fifteen
- Mid sample is tick eight, half of sixteen, away from both edges
- Step tick to watch the cursor move, or jump to mid in one click
- The verdict should show mid sample equals zero and decide equals zero
- Try eight times oversample and mid moves to tick four

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

UART oversampling

One bit cell clocked at $N\times$ **baud** (classic 16 $\times$). Find the start edge, wait to mid-bit, then sample (or majority-vote) away from the edges. Starter: 16 $\times$ cell, ideal mid sample at tick 8.

Starter example: 16 $\times$ oversampling of a clean 0 bit cell. Mid sample at tick 8 — away from the edges. Try skew and majority vote; run a start-bit center hunt.

Load starter example

Challenges 0 / 22 cleared

Quiz: oversample: 16 $\times$ oversampling means...

☐ the sample clock runs 16 times the baud rate (16 ticks per bit)

☐ there are 16 stop bits

☐ TX toggles 16 times per start bit only

☐ the FIFO depth is 16

Show hint

Check

Next

Idle

Quiz: oversample

Quiz: mid-bit

Quiz: start center

Quiz: majority

Starter

Step once

Jump to mid

Mid is 0

Select 16 $\times$

Select 8 $\times$

Bit level 1

Late skew

Mid survives skew

Majority mode

Start hunt

Demo

Explain

Sketch MID

Reset

Edge vs mid marks

Early skew

8 $\times$ mid tick

Core ideas

N $\times$ baud

Mid-bit

Start center

UART oversample starter

learn_uart — uart oversample

Real RTL/TB practice

- In Track A, restate oversampling in one sentence, how many sample clocks fit in one bit at
- Draw tick zero through fifteen for a zero bit and circle the mid sample
- Write what Start-bit hunt does after idle goes to zero
- Optional: peek at UART RX examples in the legacy combined materials and name where a
- This lab is sampling literacy, not a full synchronizer or metastability treatment yet

Pitfalls to watch

- Do not sample on the first tick after an edge, that is where skew and rise time hurt most
- Mid equals OS divided by two; at eight times that is tick four, not eight
- Majority vote helps a noisy mid but does not fix a completely wrong baud divider
- Start-bit hunt centers on the start bit
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load starter sixteen times, jump to mid at tick eight
- On paper, draw sixteen ticks and mark the mid sample for a zero bit
- When you are ready, take the short quiz, then continue to UART errors